

ICAD RT-4 · Set 10 — Answer Sheet

Exam 12-10-2026 · 75 Q · 300 marks · 3 hours · +4 correct, -1 wrong, 0 blank

Name

Roll No

Date

Physics

Chemistry

Mathematics

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Numerical — write the value

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Numerical — write the value

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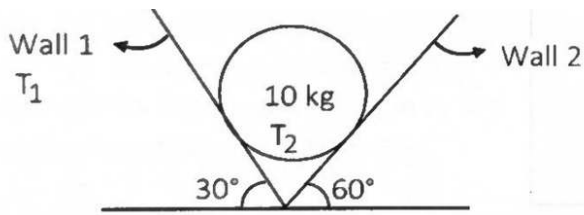
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Bubbles — single correct · Box — write the value · each subject out of 100

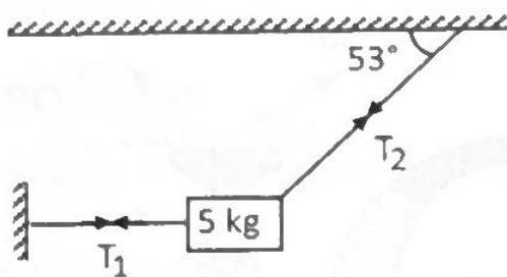
Physics

PART A · Q1–20 · SINGLE CORRECT

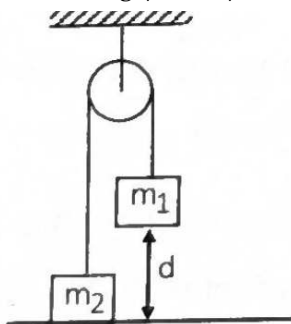
- P4L1V4Q5 A body covers one-third of the distance with velocity v_1 , the second one-third with velocity v_2 , and the remaining distance with velocity v_3 . The average velocity is: (a) $(v_1+v_2+v_3)/3$ (b) $3v_1v_2v_3/(v_1v_2+v_2v_3+v_3v_1)$ (c) $(v_1v_2+v_2v_3+v_3v_1)/3$ (d) $v_1v_2v_3/3$
- P4L9V4Q9 A body of mass m , velocity v , from a tower of height h lands at distance x . A second body of mass $2m$ from height $2h$ lands at distance $2x$. Find the horizontal velocity of the second body — (a) v (b) $2v$ (c) $\sqrt{2}v$ (d) $v/2$
- P5L2V5Q9 A 10 kg sphere rests between two walls at 30° and 60° ; the net normal force acting on the sphere by the walls is: (a) 100 N (b) 50 N (c) $100\sqrt{2}$ N (d) $50\sqrt{2}$ N



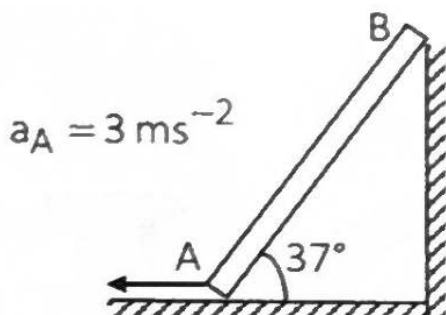
- P5L3V5Q6 A 5 kg mass hangs from a wall bracket at 53° , held by a horizontal string T_1 with T_2 along the incline. Find ratio T_2/T_1 : (a) $3/5$ (b) $3/4$ (c) $5/3$ (d) $4/3$



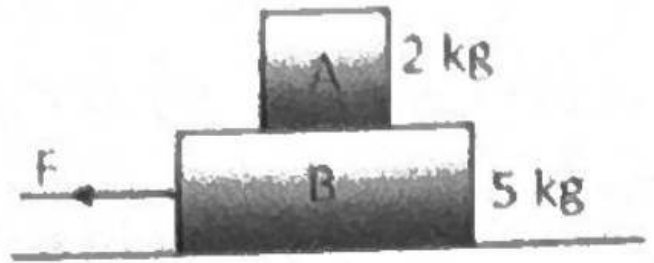
- P6L4V5Q6 If masses are released from the position shown, speed of mass m_1 just before it collides on the floor ($m_1 > m_2$). (a) $[2m_1gd/(m_1+m_2)]^{1/2}$ (b) $[2(m_1-m_2)gd/(m_1+m_2)]^{1/2}$ (c) $[2(m_1-m_2)gd/m_1]^{1/2}$ (d) None



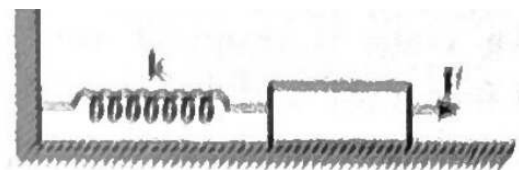
- P6L4V6Q3 Acceleration of point B at the instant shown (rod remains in contact with wall and floor), given $a_A = 3 \text{ m/s}^2$, angle 37° . (a) 3 (b) 5 (c) 4 (d) 7 m/s^2



- P7L2V5Q3 A block of mass m is placed at rest on a horizontal rough surface with angle of friction ϕ . The block is pulled with a force F at an angle θ with the horizontal. The minimum value of F required to move the block is: (a) $mg \sin\phi/\cos(\theta - \phi)$ (b) $mg \cos\phi/\cos(\theta - \phi)$ (c) $mg \tan\phi$ (d) $mg \sin\phi$
- P7L2V6Q4 A body of mass M is kept on a rough horizontal surface (friction coefficient $= \mu$). A person is trying to pull the body by applying a horizontal force but the body is not moving. The force by the surface on the body is F , where: (a) $F = Mg$ (b) $F = \mu Mg$ (c) $Mg \leq F \leq Mg\sqrt{1 + \mu^2}$ (d) $Mg \geq F \geq Mg\sqrt{1 - \mu^2}$
- P7L4V3Q3 Two blocks A (2 kg) and B (5 kg) rest one over the other on a smooth horizontal plane. The coefficient of static and dynamic friction between A and B is the same and equal to 0.60. The maximum horizontal force that can be applied to B so that A and B have no relative motion is: (a) 4.2 N (b) 42 N (c) 5.4 N (d) 1.2 N



- P8L2V4Q6 A force $F = a + bx$ acts on a particle in the x -direction where a and b are constants. Find the workdone by this force during a displacement from $x = 0$ and $x = d$? (a) $(a + (1/2)bd)d$ (b) $(ad + (1/2)bd)$ (c) $(ad + bd)/2$ (d) $bd^2/2$
- P8L3V5Q4 A block attached to a spring, pulled by a constant horizontal force, is kept on a smooth surface as shown in the figure. Initially, the spring is in the natural state. Then, the maximum positive work that the applied force F can do is [given that, spring does not break] (a) F^2/k (b) $2F^2/k$ (c) $3F^2/2k$ (d) $F^2/2k$

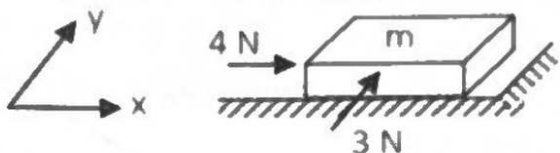


- P8L4V2Q2 A 250 g block slides on a rough horizontal table. Find the work done (in J) by the frictional force in bringing the block to rest if it is initially moving at a speed of 40 cm/s. (write the value)
- P8L4V5Q4 A particle moving along the x axis is acted upon by a single force $F = F_0e^{-kx}$, where F_0 and k are constants. The particle is released from rest at $x = 0$. It will attain a maximum kinetic energy of: (a) F_0/k (b) $F_0/e k$ (c) kF_0 (d) $1/2(kF_0)^2$
- P8L7V3Q4 A vertical spring with force constant k is fixed on a table. A ball of mass m at a height h above the free upper end of the spring falls vertically on the spring so that the spring is compressed by a distance d . The net work done in the process is: (a) $mg(h+d) - (1/2)kd^2$ (b) $mg(h-d) - (1/2)kd^2$ (c) $mg(h-d) + (1/2)kd^2$ (d) $mg(h+d) + (1/2)kd^2$

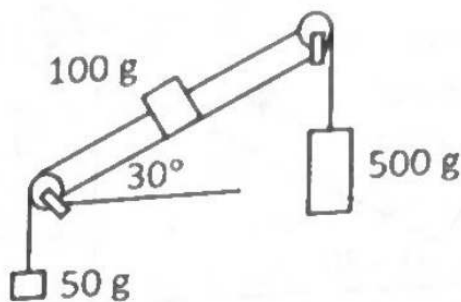
15. P4L4V5Q4 A ball is thrown upwards from the ground. It is at a height of 100 m in the upward and downward journeys at times t_1 and t_2 respectively. If $g=10 \text{ m/s}^2$, then $t_1 t_2$ is equal to: (a) 10 (b) 20 (c) 40 (d) 50
16. P2L5V3Q3 $\int ((4-x)/x) dx$. (a) $4 \ln x + c$ (b) $4 \ln x - 2x + c$ (c) $4 \ln x - x + c$ (d) $\ln x - x + c$
17. P3L6L7V4Q10 Given $A=2\hat{i}+p\hat{j}+q\hat{k}$ and $B=5\hat{i}+7\hat{j}+3\hat{k}$. If $A \parallel B$, the values of p and q are respectively: (a) $14/5$ and $6/5$ (b) $14/3$ and $6/5$ (c) $6/5$ and $1/3$ (d) $3/4$ and $1/4$

PART B · Q21–25 · NUMERICAL VALUE

21. P6L2V3Q1 Resultant force on block: 4 N ($\rightarrow$) and 3 N ($\uparrow$) perpendicular



22. P6L3V5Q2 Acceleration of the 500 g block; if the answer = $8g/x$, find x (100 g on a 30° incline, 500 g hanging, 50 g).

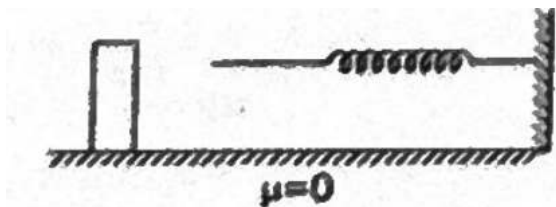


23. P7L3V5Q4 A block of mass 2.5 kg is kept on a rough horizontal surface. It is found that the block does not slide if a horizontal force less than 15 N is applied to it. It is also found that it takes 5 seconds to slide through the first 10 m if a horizontal force of 15 N is applied and the block is gently pushed to start the motion. Taking $g = 10 \text{ m/s}^2$, calculate the sum of the coefficients of static and kinetic friction between the block and the surface.

Chemistry

PART A · Q1–20 · SINGLE CORRECT

1. C2L6V3Q4 Compute ratio v_2 / v_5 for a hydrogen electron. (a) 2.5 (b) 0.4 (c) 1.25 (d) 3.5
2. C2L11V3Q1 Calculate radial and angular nodes for a 4p orbital. (a) 2, 1 (b) 3, 1 (c) 2, 2 (d) 1, 2
3. C2L11V5Q5 A_n = area of n th H-atom orbit; graph of $\ln(A_n/A_1)$ vs $\ln n$. (one or more options may be correct). (a) will pass through origin (b) will be straight line with slope = 4 (c) will be monotonically increasing non-linear curve (d) will be a circle
4. C3L2V6Q8 1 mole of $^{14}\text{N}^{3-}$ ions contains: (a) $7 \times 6.023 \times 10^{23}$ electrons (b) $7 \times 6.023 \times 10^{23}$ protons (c) $7 \times 6.023 \times 10^{23}$ neutrons (d) $14 \times 6.023 \times 10^{23}$ protons
5. C3L10V3Q1 Calculate molality of 18 g glucose ($M = 180 \text{ g/mol}$) in 0.5 kg water. (a) 0.05 m (b) 0.1 m (c) 0.2 m (d) 0.3 m
6. C3L10V3Q4 18 g glucose ($M = 180$) is dissolved in 90 g water ($M = 18$). Find mole fraction of solute. (a) 0.02 (b) 0.1 (c) 0.2 (d) 0.15
7. C3L11V4Q8 How many significant figures are present in 0.010100×10^3 ? (a) 7 (b) 5 (c) 3 (d) 10
18. P3L6L7V4Q4 Find angular momentum $L = r \times p$ where $r = 2\hat{i} + \hat{j}$ and $p = 3\hat{j} + 2\hat{k}$. (a) $2\hat{i} - 4\hat{j} + 6\hat{k}$ (b) $6\hat{i} - 4\hat{j} + 2\hat{k}$ (c) $-2\hat{i} + 4\hat{j} - 6\hat{k}$ (d) $4\hat{i} - 6\hat{j} + 2\hat{k}$
19. P4L8V2Q4 Body projected with speed u at angle β . Kinetic energy at highest point is $3/4$ of initial KE. Value of β ? (a) 30° (b) 45° (c) 60° (d) 120°
20. P4L10V5Q8 A police car (speed v) is passed by a truck (speed $3v/2$) at $t = 0$. The police car starts accelerating at 3 m/s^2 10 s later, and takes a further 10 s to catch the truck. Find v . (a) 10 m/s (b) 15 m/s (c) 20 m/s (d) 30 m/s
24. P8L1V6Q9 A block A of mass 10 kg is placed over a long plank B of mass 20 kg. The system rests on a smooth horizontal surface. The coefficient of friction between the blocks is 0.5. A horizontal constant force of 60 N is applied on the plank B. (take $g = 10 \text{ m/s}^2$) i. If block A and plank B are taken as a system, find the work done by the friction on the system in 2 s. ii. If block A and plank B are taken as a system, the net work done on the system in 2 s is 80x J. Find the value of x .
25. P8L3V6Q7 A block of mass m kg moving with kinetic energy 8 J on a smooth horizontal surface hits the free end of a spring, whose other end is attached to a rigid wall as shown in the figure. If the restoring force applied by the spring is given by $F(x) = (-2x - x^3) \text{ N}$, then find the maximum compression (in m) in the spring.



13. C5L5V5Q9 In which species the hybrid state of central atom is/are sp^3d ? (a) $I_3(-)$ (b) SF_4 (c) PF_5 (d) IF_5
14. C5L7V3Q5 Why does head-on overlap of atomic orbitals form a stronger bond than sidewise overlap? (a) It forms π -bond (b) It has minimum overlap region (c) It produces symmetrical electron density along internuclear axis (d) It leads to non-bonding interaction
15. C5L7V4Q1 Antibonding molecular orbital is formed by: (a) addition of wave functions of atomic orbitals (b) subtraction of wave functions of atomic orbitals (c) multiplication of wave functions of atomic orbitals (d) none of these
16. C2L10V5Q8 Ground-state Cr ($Z=24$) — total orbitals populated by ≥ 1 electron. (a) 15 (b) 16 (c) 20 (d) 14
17. C5L4V5Q8 How many of the following molecule are sp^3d or dsp^3 hybridized: SF_4 , SCl_4 , XeO_2F_2 , CH_4 , PCl_5 , PCl_3 , XeO_3 (a) 2 (b) 3 (c) 4 (d) 5
18. C5L8V5Q3 Bond order of N_2^+ , N_2^- and N_2 will be (a) 2.5, 2.5 and 3 respectively (b) 2, 2.5 and 3 respectively (c) 3, 2.5 and 3 respectively (d) 2.5, 2.5 and 2.5 respectively
19. C2L3V4Q10 Which of the following statements are correct?
Statement I: Increasing the intensity of incident light increases the number of photoelectrons emitted per second.
Statement II: Increasing the intensity of light increases the maximum kinetic energy of emitted photoelectrons.
Statement III: Maximum kinetic energy of photoelectrons depends only on the frequency of incident light. (a) I and II only (b) II and III only (c) I and III only (d) I, II and III
20. C1L3V6Q1 12 g ideal gas ($M=120$), 1.00 atm at $t^\circ C$; $+10^\circ C \rightarrow 1.10$ atm — find t . (a) $-273^\circ C$ (b) $-173^\circ C$ (c) $-100^\circ C$ (d) $100^\circ C$

PART B · Q21–25 · NUMERICAL VALUE

21. C3L11V6Q6 A student performs a titration with different burettes and finds titre values of 25.2 mL, 25.25 mL and 25.0 mL. The number of significant figures in the average titre value is ...
22. C1L1V4Q9 Barometer 750, open manometer gas 40 mm higher — vs 1 bar. (a) $P_{\text{gas}} < 1$ bar (b) $P_{\text{gas}} = 1$ bar (c) $P_{\text{gas}} > 1$ bar (d) Need Hg density to decide
23. C1L2V5Q3 1.0 atm, 3.0 L, 300 K — compress to half volume & heat to 600 K; final pressure. (a) 2.0 (b) 3.0 (c) 4.0 (d) 5.0
24. C1L3V4Q9 1000 mL gas at $-73^\circ C$ & 2.0 atm $\rightarrow 127^\circ C$ & 0.5 atm — final volume. (a) 4000 mL (b) 6000 mL (c) 8000 mL (d) 10000 mL
25. C1L10V6Q8 Assertion: Surface tension decreases with rise in temperature and becomes zero at the critical temperature. Reason: With increase in temperature, cohesive forces between molecules decrease. (a) Both A and R true, R explains A (b) Both A and R true, R does not explain A (c) A true, R false (d) A false, R true

Mathematics

PART A · Q1–20 · SINGLE CORRECT

1. M5AL1V5Q3 If $3\pi/2 < \theta < 2\pi$, then $\sqrt{((1 + \cos\theta)/(1 - \cos\theta))}$ is equal to: (a) $\operatorname{cosec}\theta + \cot\theta$ (b) $\operatorname{cosec}\theta - \cot\theta$ (c) $-\operatorname{cosec}\theta + \cot\theta$ (d) $-\operatorname{cosec}\theta - \cot\theta$
2. M5BL1V3Q1 If $\cos 72^\circ - \cos 48^\circ = k \sin 12^\circ$, then $k = ?$ (a) $\sqrt{3}$ (b) $-\sqrt{3}$ (c) $-1/\sqrt{3}$ (d) $1/\sqrt{3}$
3. M6L2V6Q7 $(x + 1)/(x - 1)^2 < 1$, then 'x' belongs to: (a) $(-\infty, 0) \cup (3, \infty)$ (b) $(-\infty, -1) \cup (2, \infty)$ (c) $(-\infty, -2) \cup (4, \infty)$ (d) none of these
4. M6L3V5Q9 For $t \in \mathbb{R}$, consider $|x - 1| + |x + 1| = t$. (One or more options may be correct.) (a) No solution for $t < 2$ (b) Infinitely many solutions for $t = 2$ (c) Exactly two solutions for each $t > 2$ (d) Exactly one solution for some $t > 2$
5. M7L2V5Q5 Evaluate $\log_{\sqrt{3}}(81) + \log_{\sqrt{5}}(1/4\sqrt{5})$ ($\sqrt{5}$)⁷ (a) -19 (b) -26 (c) -15 (d) -22
6. M7L2V5Q7 Number of ordered pair(s) satisfying simultaneously the system $2^{(\sqrt{x} + \sqrt{y})} = 256$ and $\log_{10} \sqrt{xy} - \log_{10} 1.5 = 1$, is (a) zero (b) exactly one (c) exactly two (d) more than two
7. M8AL2V5Q8 Three numbers are in A.P. whose sum is 33 and product is 792, then the smallest number from these number is (a) 4 (b) 8 (c) 11 (d) 14
8. M8BL1V5Q6 If a_1, a_2, a_3 ($a_1 > 0$) are three successive terms of a G.P. with common ratio 'r' the value of 'r' for which $a_3 > 4a_2 - 3a_1$ holds is given by (a) $1 < r < 3$ (b) $-3 < r < -1$ (c) $r > 3$ or $r < -1$ (d) None of these
9. M8BL3V4Q5 If a, b, c are in G.P., then (a) a^2, b^2, c^2 are in G.P. (b) $a^2(b+c), c^2(a+b), b^2(a+c)$ are in G.P. (c) $a/(b+c), b/(c+a), d/(a+b)$ are in G.P. (d) None of these
10. M8BL3V5Q3 If a, b, c are in A.P. then $2^{(ax+1)}, 2^{(bx+1)}, 2^{(cx+1)}$, $x \neq 0$ are in (a) A.P. (b) G.P. when $x > 0$ (c) G.P. when $x < 0$ (d) G.P. $\forall x$
11. M8CL1V5Q9 If a, b, c are real and in A.P. and a^2, b^2, c^2 are in H.P., then (a) $a = b = c$ (b) $2b = 3a + c$ (c) $b^2 = \sqrt{ac/8}$ (d) none of these
12. M5CL3V6Q1 If $(2^n + 1)\theta = \pi$, then $2^n \cdot \cos\theta \cdot \cos 2\theta \cdot \cos 2^2\theta \dots \cdot \cos(2^{n-1}\theta) =$ (a) -1 (b) 1 (c) 1/2 (d) none
13. M7L3V6Q3 Given $\log_2(\log_8 x) = \log_8(\log_2 x)^2$, then the value of $\log_2 x$ equal to: (a) 9 (b) 12 (c) 27 (d) $3\sqrt{3}$
14. M8AL1V5Q7 An A.P. consists of 23 terms. If the sum of the three terms in the middle is 141 and the sum of the last three terms is 261, then the first term is (a) 6 (b) 5 (c) 4 (d) 3
15. M8BL2V5Q2 If there are two infinite GP 6, 4, $8/3$, and $\lambda, \lambda/2, \lambda/4$, If sum of the sum of these two infinite GP is 20 then the value of λ is (write the value)
16. M8CL1V6Q1 If $a = 1 + 1/2 + 1/4 + 1/8 + \dots \infty$ terms and $b = 1 + 1/3 + 1/9 + 1/27 + \dots \infty$ terms, and if a, c, b are in H.P., then the value of $7c$ is (write the value)
17. M1L1V5Q3 A point moves along a circle of radius 7 cm from an initial position P through an angle $\theta = \pi/3$ to reach Q. If the point instead traverses $\theta' = \pi/3 + 6\pi$, what is the net difference in arc lengths measured along the same (positive) direction from P to Q' and from P to Q? (a) 84π cm (b) 42π cm (c) 14π cm (d) 0
18. M2L1V5Q1 In a group of 500 people, 200 can speak Hindi alone while only 125 can speak English alone. The number of people who can speak both Hindi and English is. (a) 175 (b) 325 (c) 300 (d) 375
19. M4L3V4Q1 Let $G(x) = \{1/x\}$, the fractional part of $1/x$, for x in $(0, 1]$. Which of the following is/are true? (one or more options may be correct). (a) $G(x)$ takes every value in $(0, 1)$ infinitely often. (b) $\lim_{x \rightarrow 0^+} G(x)$ exists. (c) $G(x) = 0$ for infinitely many x . (d) $G(x)$ is periodic

- 20.** M4L3V4Q4 For real x in $[0,10)$, define $F(x) = \text{floor}(\{x\} + x/3)$. Which of the following sets equals the range of F on $[0,10)$?
(a) $\{0,1,2,3\}$ (b) $\{0,1,2,3,4\}$ (c) $\{1,2,3,4\}$ (d) $\{0,1,2,3,4,5\}$

PART B · Q21–25 · NUMERICAL VALUE

- 21.** M5CL2V4Q3 If $\cos x = -3/5$ and x lies in II quadrant, then the value of $\sin(x/2)$ is _____
- 22.** M6L5V3Q4 If Domain of $f(x) = (2x + 1)/(x^2 - 3)$ is $\mathbb{R} - \{x_1, x_2\}$ then $x_1 + x_2$ is:
- 23.** M8AL1V5Q1 The 4th term of an A.P. is three times the first, and the 7th term exceeds twice the third term by 1. Find the sum of the 1st term and the common difference of this A.P.
- 24.** M8AL3V3Q2 If the numbers a, b, c, d, e form an A.P. with $a = 1$, then $a - 4b + 6c - 4d + e =$
- 25.** M3L3V4Q1 $6\sin^2\theta - 11\sin\theta + 4 = 0 \rightarrow$ find $\sin\theta$